BhR2 bmr6717 largest child

11 AIS methods · 9,856 selected models

Methodology

Sampling methods analyzed

11 AIS methods; 9,856 selected models.
AFSample2; AFSample3; Boltz2; OpenDDE.
BioEMU; BioEMU2; AFCluster; SPEACH_AF.
AlphaFlow MD; AlphaFlow PDB; ESMFlow PDB.

Shared structural feature input

Second pass: only primary Cluster 1 models.
Use the frozen parent residue/pair mask.
107 retained residues; 5,671 Cα distance pairs.
Fresh PCA, UMAP and t-SNE fit this subset.
Primary labels and native structures are preserved.
Final child-count review remains open.

Two-dimensional projections

PCA: scikit-learn 1.7.1; centered SVD.
t-SNE: scikit-learn 1.7.1; Barnes–Hut,
perplexity 30; the workflow uses Python t-SNE.
UMAP: umap-learn 0.5.11; neighbors 15,
min_dist 0.1; Euclidean distance. Random seed 42.

Clusters and displayed representatives

Preserve UMAP components; seeds 42/7/123.
Ward splits: feature silhouette ≥0.15; ARI ≥0.80.
Up to 5 models per cluster by mean pLDDT.
7,858 selected models have native confidence.
BioEMU/BioEMU2 stay unscored in clustering.
Unscored-only clusters use structural medoids.

Medoid TM-similarity grid

Among the displayed models in each cluster, the medoid minimizes mean pairwise aligned Cα RMSD over shared, chain-corresponded residues.
US-align scores each medoid pair in both directions; average-length normalization and directional mean give a symmetric grid (0–1; higher is more similar).

Bonsai of the selected ensembles

Bonsai / Bonsai Scout: native engine b16ef6f.
16 representatives; 15 nonzero PCA axes.
Variance scaling off; tree paths encode distance.
Native edge cuts select 2 groups by silhouette.
B labels apply only to selected representatives.
Circular views and dendrogram; root is arbitrary.

PCA / scikit-learn: Pedregosa et al., 2011

t-SNE: van der Maaten & Hinton, 2008

UMAP: McInnes, Healy & Melville, 2018

Only nine confidence-producing methods are ranked; BioEMU/BioEMU2 remain in every structural analysis. Unscored-only clusters retain their structural medoid.

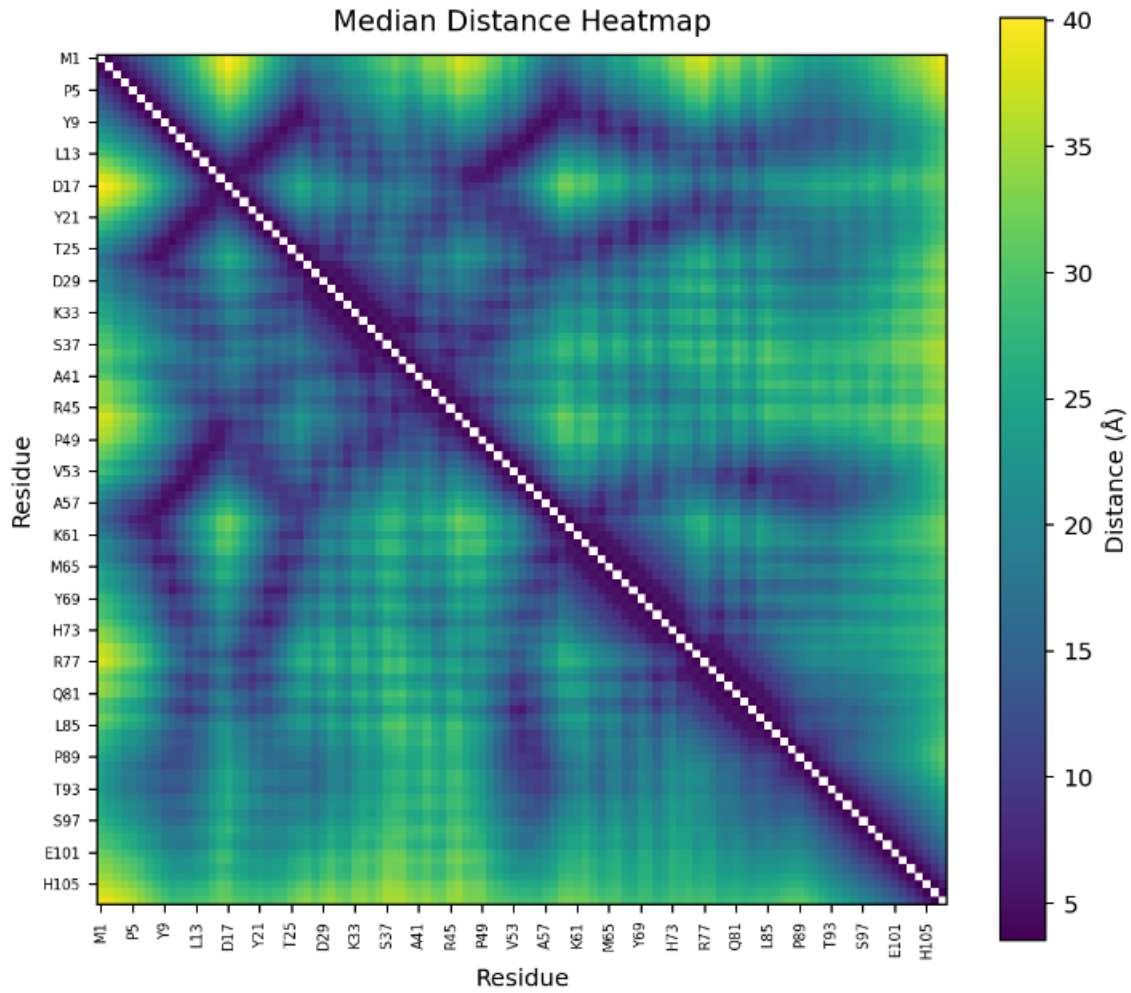


Figure 1. Median Cα distance between shared residues across the model ensemble.

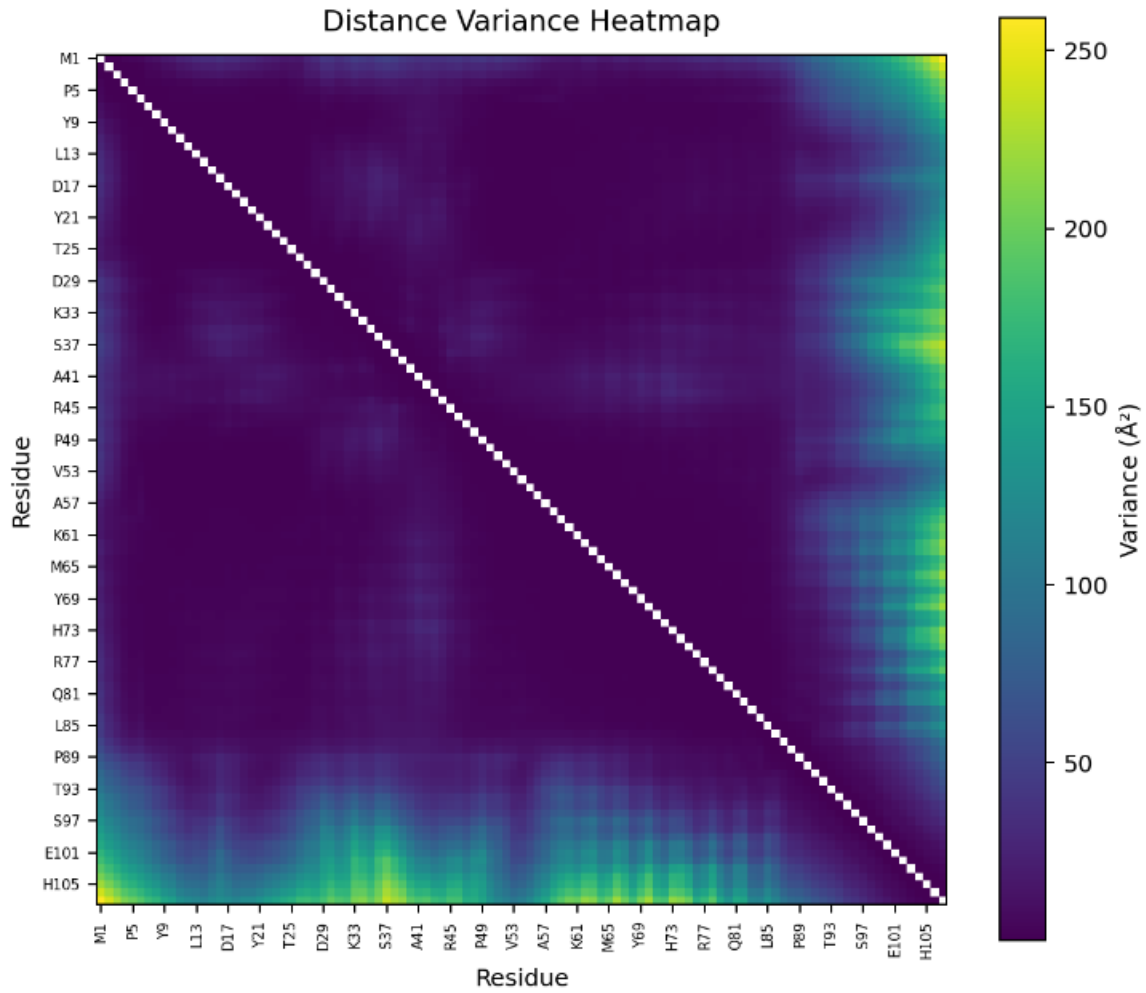


Figure 2. Variance of Cα distances between shared residues across the model ensemble.

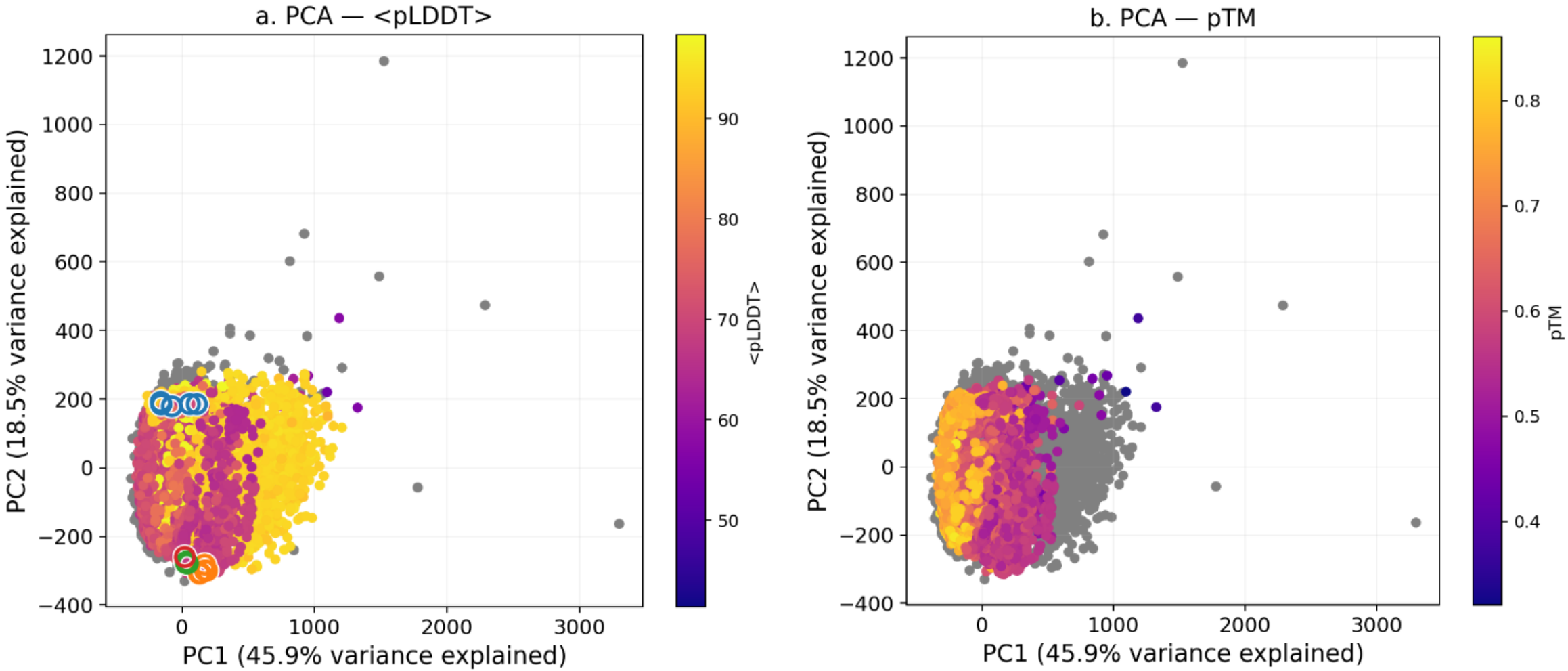


Figure 3. PCA embedding of shared-residue distograms, colored by confidence metrics. a: <pLDDT>; b: pTM. Cluster-colored outlines in panel a identify the selected representative models from each structure cluster ensemble. Gray models do not have this value calculable.

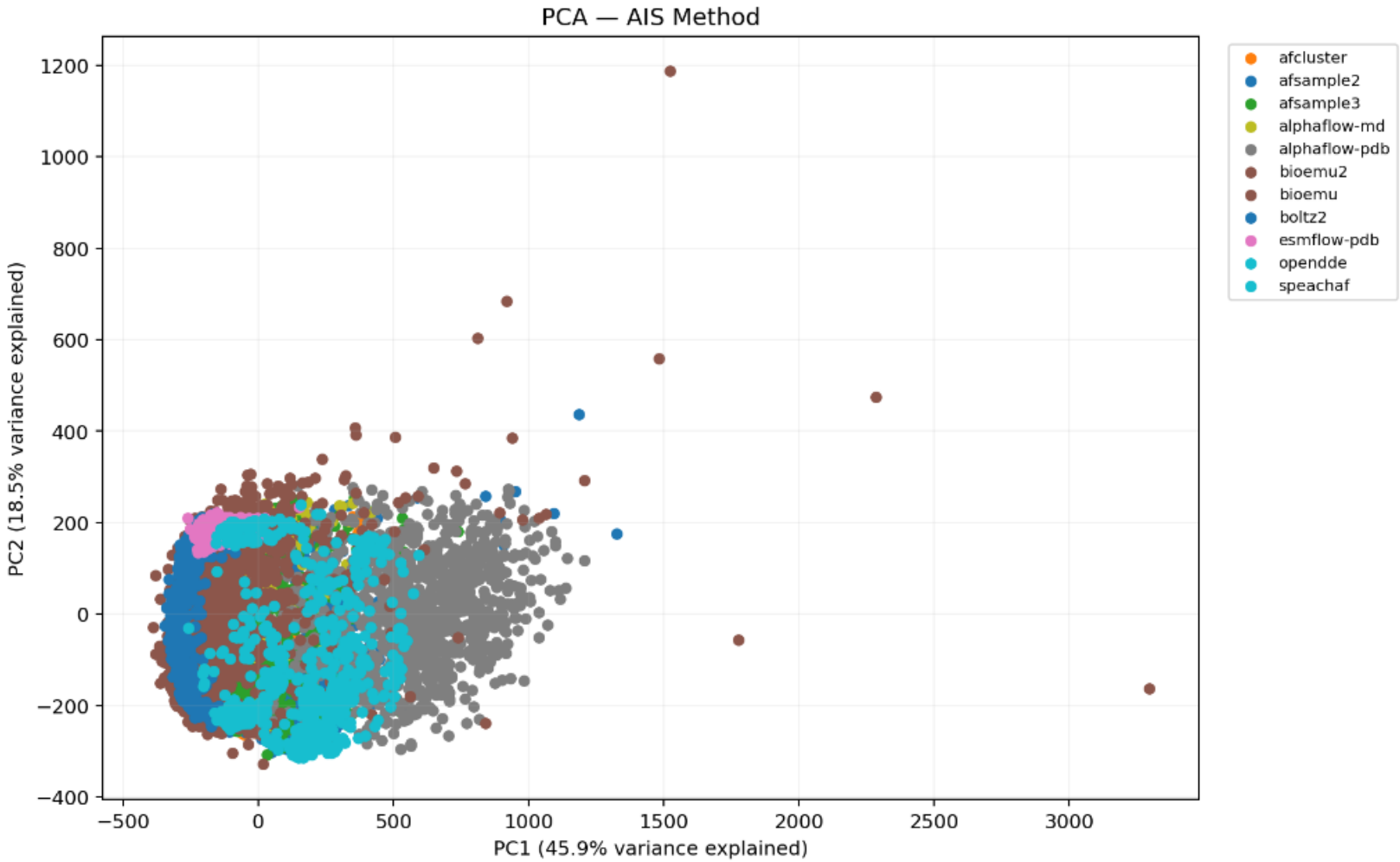


Figure 4. PCA embedding of shared-residue distograms, colored by AIS Method.

a. Cluster labels

b. Model types

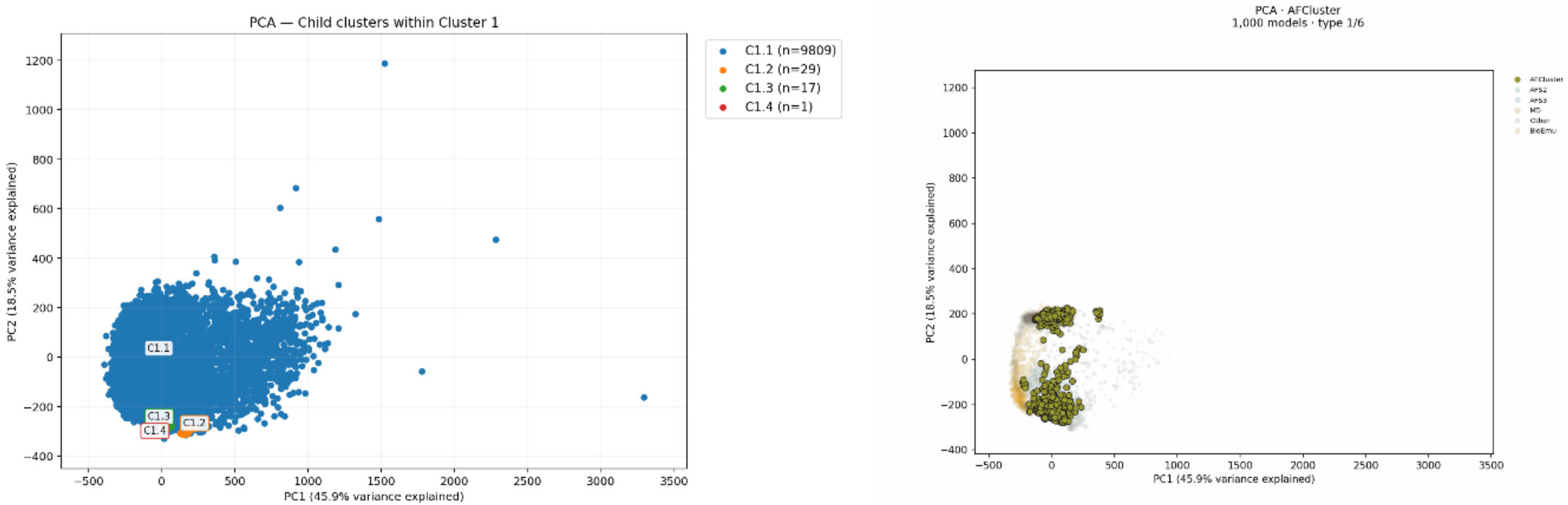


Figure 5. PCA projection of shared-residue distograms. a: cluster labels assigned from UMAP; cluster colors are consistent across projections. b: model type labels; each model type uses its consistent color across workflow runs.

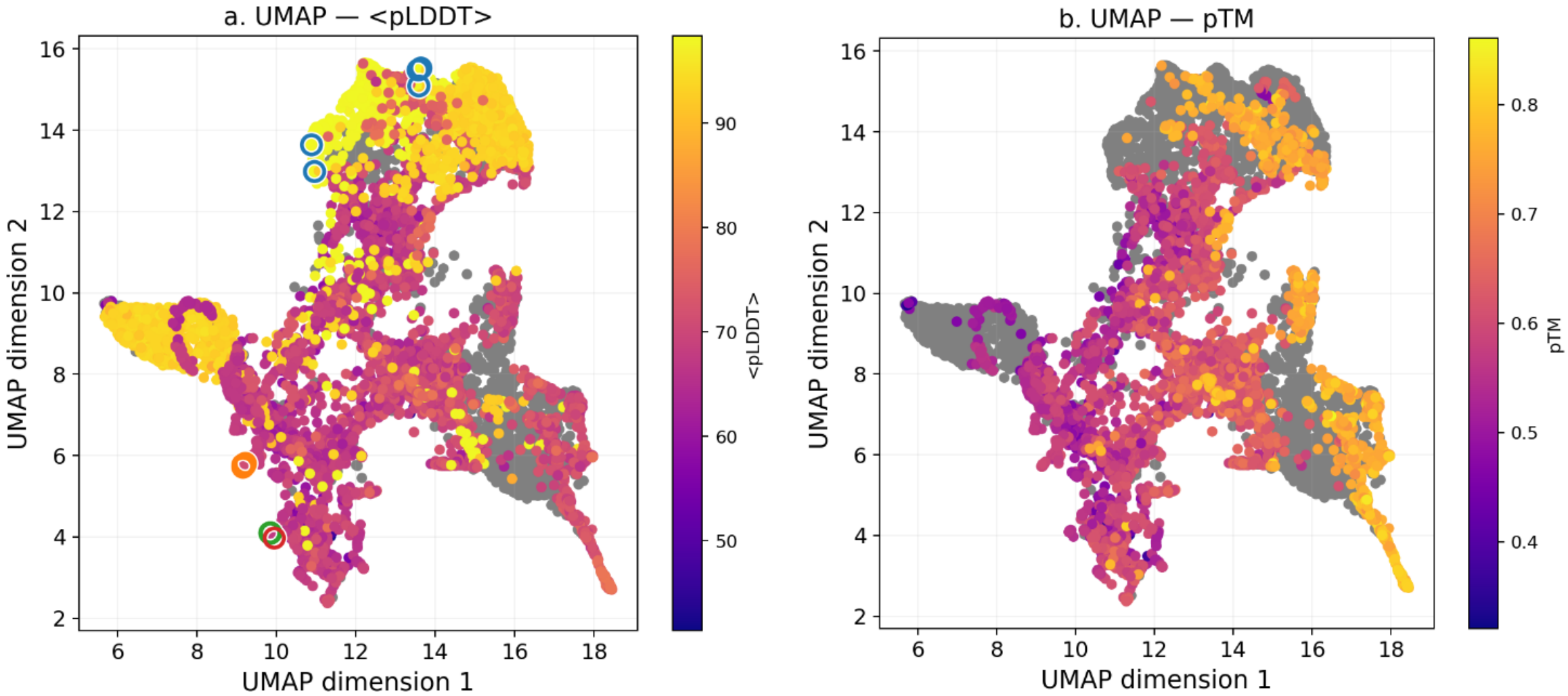


Figure 6. UMAP embedding of shared-residue distograms, colored by confidence metrics. a: <pLDDT>; b: pTM. Cluster-colored outlines in panel a identify the selected representative models from each structure cluster ensemble. Gray models do not have this value calculable.

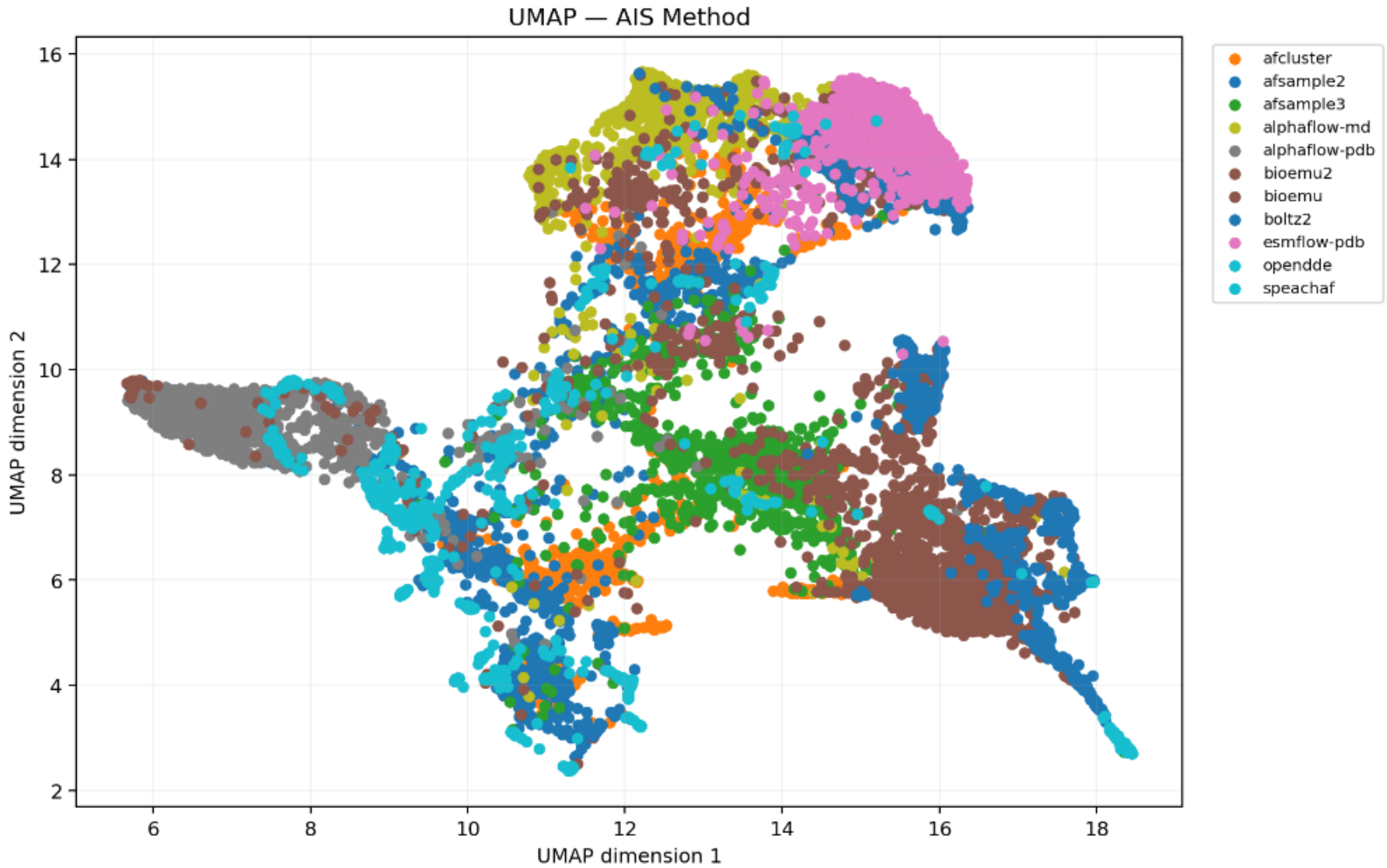


Figure 7. UMAP embedding of shared-residue distograms, colored by AIS Method.

a. Cluster labels

b. Model types

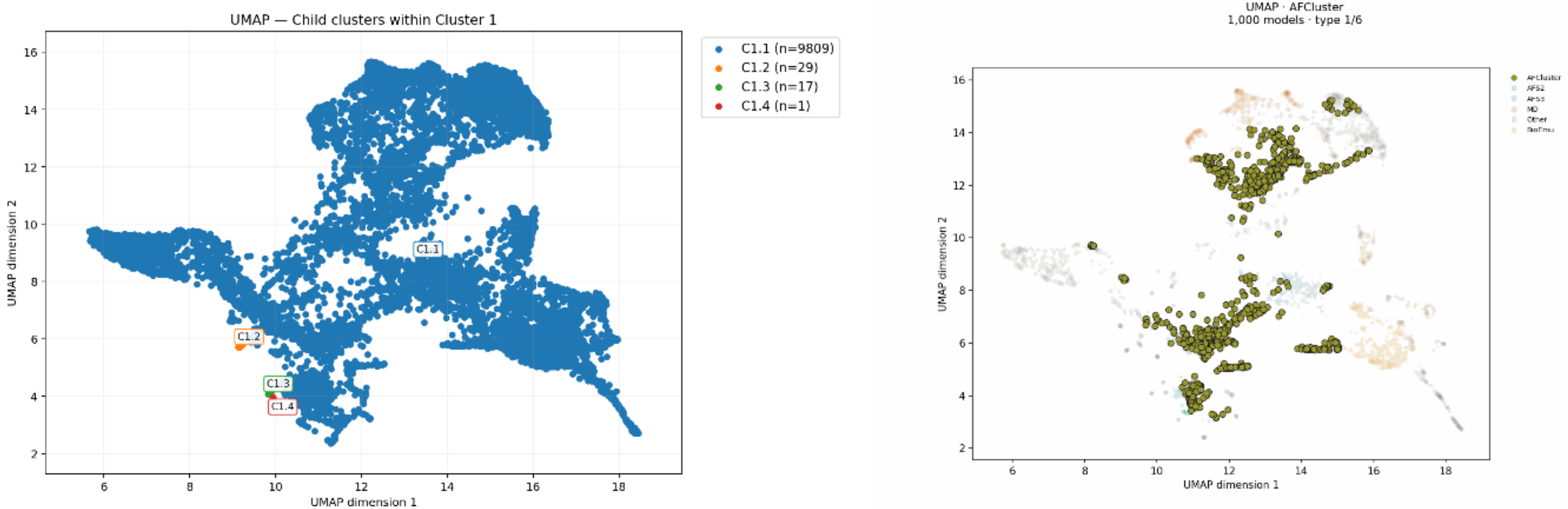


Figure 8. UMAP projection of shared-residue distograms. a: cluster labels assigned from UMAP; cluster colors are consistent across projections. b: model type labels; each model type uses its consistent color across workflow runs.

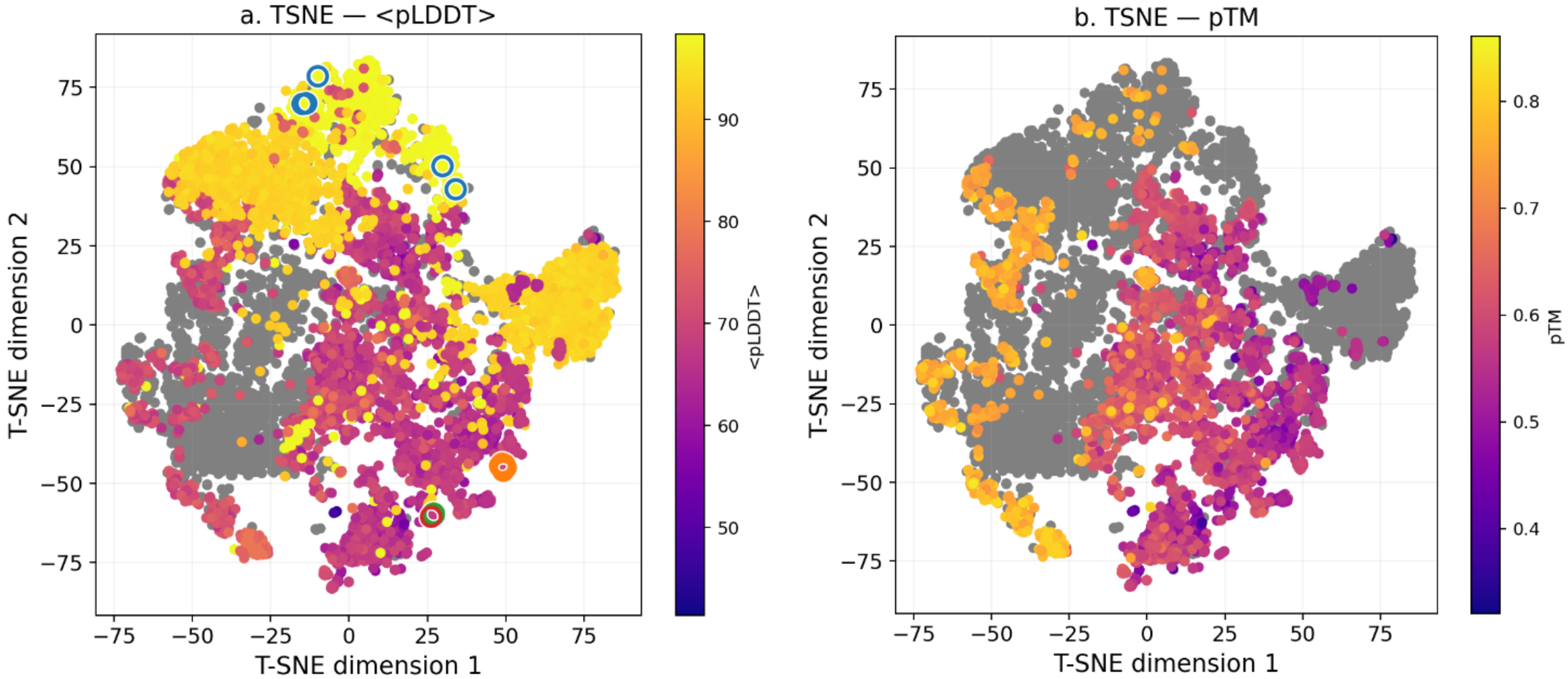


Figure 9. t-SNE embedding of shared-residue distograms, colored by confidence metrics. a: <pLDDT>; b: pTM. Cluster-colored outlines in panel a identify the selected representative models from each structure cluster ensemble. Gray models do not have this value calculable.

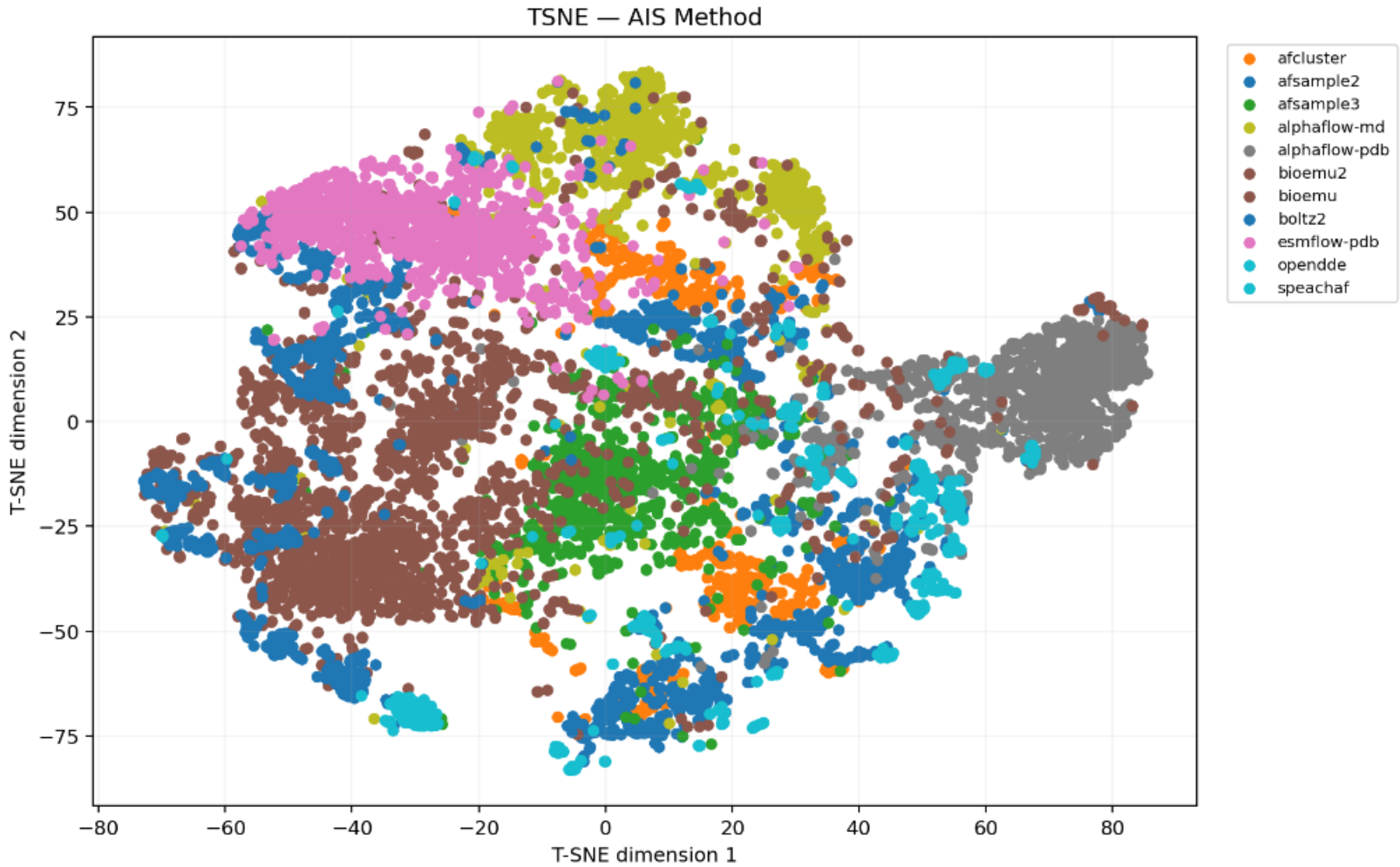
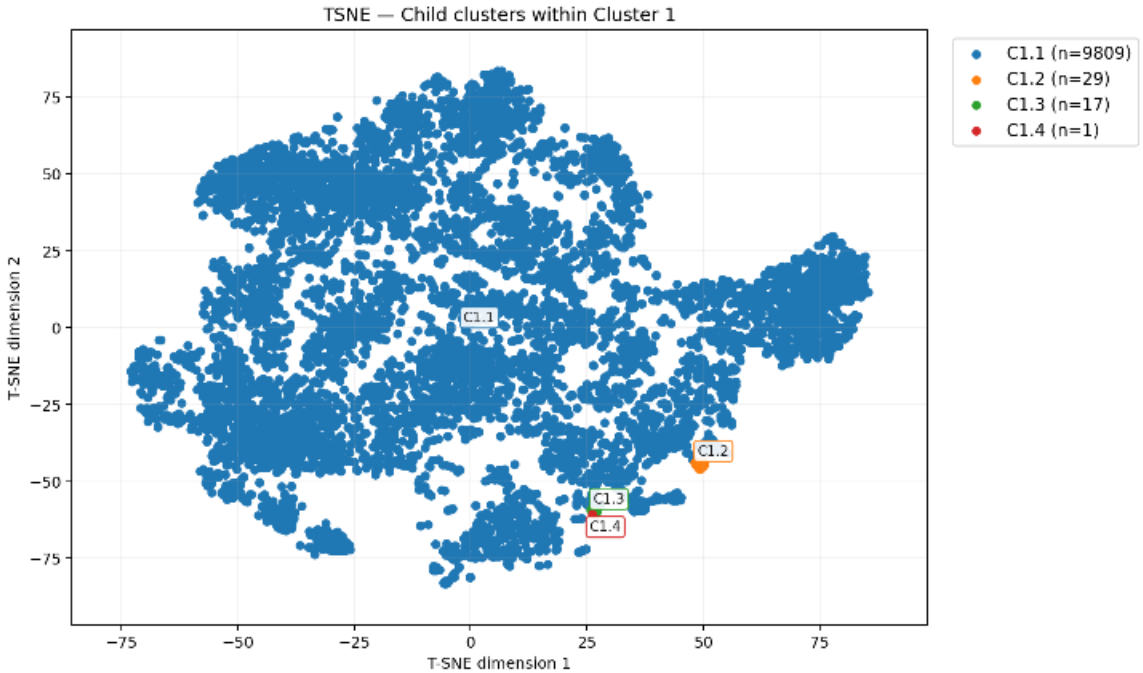


Figure 10. t-SNE embedding of shared-residue distograms, colored by AIS Method.

a. Cluster labels



b. Model types

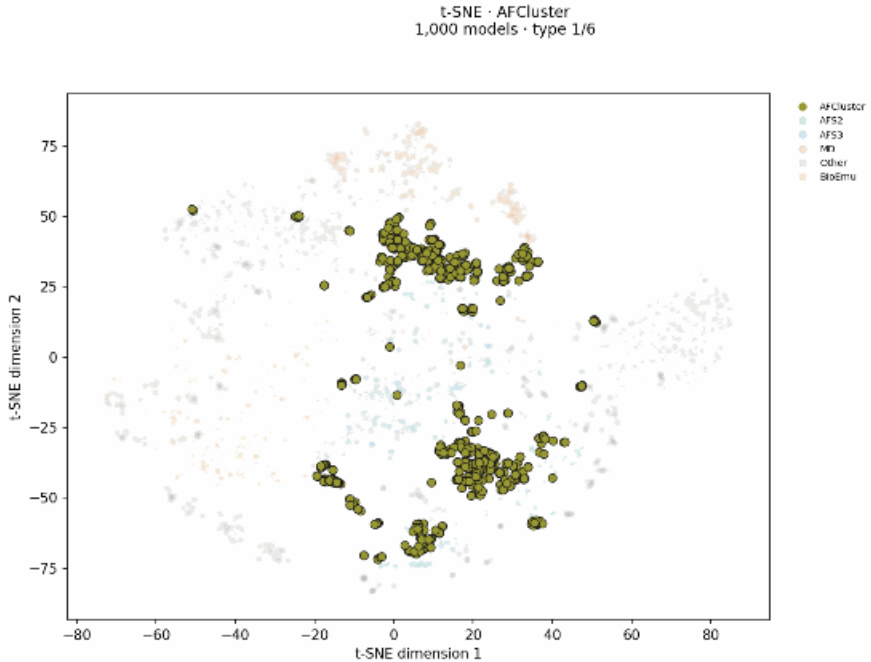
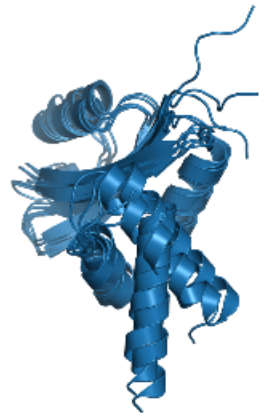


Figure 11. t-SNE projection of shared-residue distograms. a: cluster labels assigned from UMAP; cluster colors are consistent across projections. b: model type labels; each model type uses its consistent color across workflow runs.

a. C1.1



b. C1.2



c. C1.3



d. C1.4

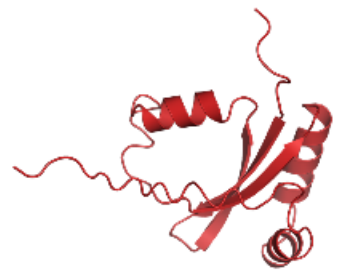


Figure 12. Cluster ensembles (up to 5 models each), aligned to each full-cluster Cα RMSD medoid. Eligible models ranked by <pLDDT>; representative selection follows the recorded source scope.

Cluster confidence distributions

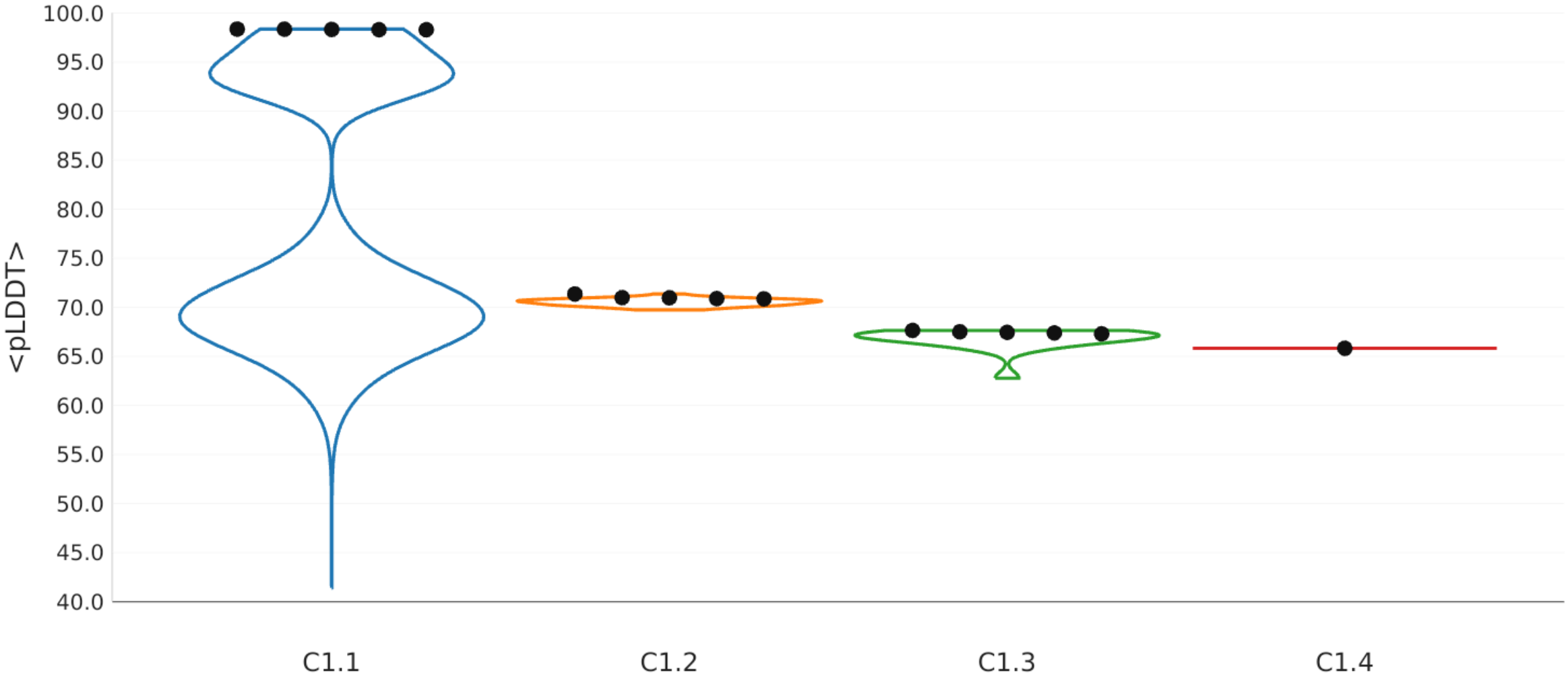
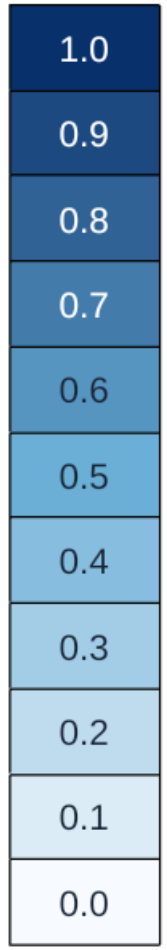


Figure 13. Cluster confidence distributions. Colored outlines show eligible cluster members with confidence scores; black dots mark the displayed models. Eligible models ranked by <pLDDT>; representative selection follows the recorded source scope.

Selected-model medoid TM-scores

Rows and columns identify the same clusters

	C1.1	C1.2	C1.3	C1.4
C1.1	1.00	0.65	0.62	0.60
C1.2	0.65	1.00	0.78	0.77
C1.3	0.62	0.78	1.00	0.86
C1.4	0.60	0.77	0.86	1.00



Cluster groups

Group 1: C1.3, C1.4

All pairs ≥ 0.80
Groups may overlap

Figure 14. Whole-chain TM-scores between medoids of the selected models in each cluster.
Average-length normalization, mean of both directions. Higher scores indicate greater structural similarity.

Bonsai · selected cluster ensembles (native engine b16ef6f)

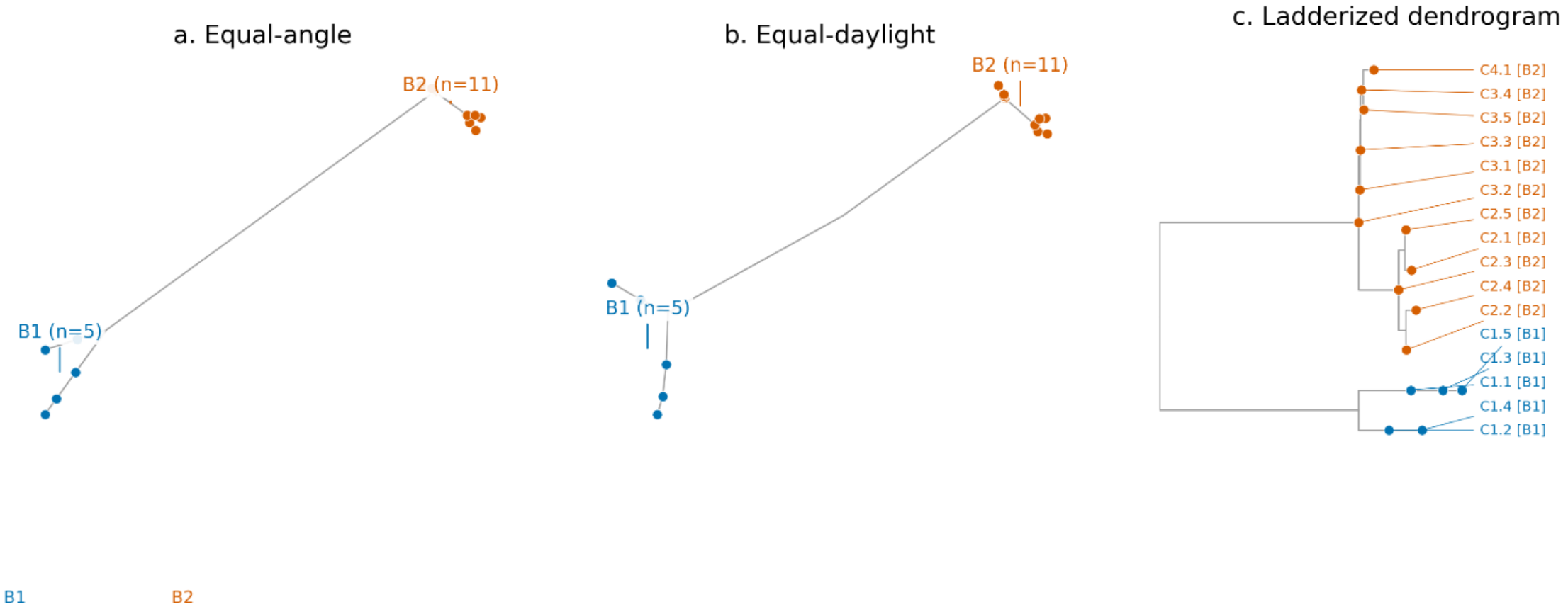


Figure 15. Bonsai of 16 representatives; 15 nonzero PCs, variance scaling off. a: equal-angle; b: equal-daylight; c: dendrogram. Bonsai-derived groups: native edge cuts, k=2 (tree-distance silhouette 0.896). Cx.y retains original identity; B groups apply only to selected models. Other slides retain UMAP labels. Paths encode distance; root is arbitrary.

TM-similar cluster ensembles

a. Group 1: C1.3, C1.4



Figure 16. Maximal groups whose every cluster-medoid pair has TM-score > 0.8 (group 1: 6 models, TM 0.8641–0.8641). All displayed representatives are pooled and aligned to the pooled Cα RMSD medoid using the recorded alignment residues, without outlier rejection. Detected terminal exclusions also apply to fitting. Colors match PCA, UMAP and t-SNE clusters; saved camera orientation is reused. Groups may overlap.

Pooled group 1 · one model per frame

Whole-structure fit



Complete structures, one model per frame. Same fit and fixed camera as the corresponding ensemble slide; each chain repeats its own N-to-C rainbow. Maximum-RMSD endpoints with minimum-transition path search. Five frames at 30 fps per model (~0.167 s). GIF loop reset is excluded from the path objective. The PDF shows the first frame.